7. It is given that 5% of the electric bulbs manufactured by a company are defective. Using poisson distribution find the probability that a sample of 120 bulbs will contain no defective bulb.
8. A car hiring firm has two cars. The demand for cars on each day is distributed as a Poisson variate, with mean 1.5. Calculate the proportion of days on which
   (i) Neither car is used
   (ii) Some demand is refused
9. The average number of phone calls per minute into the switch board of a company between 10.00 am and 2.30 pm is 2.5. Find the probability that during one particular minute there will be (i) no phone at all (ii) exactly 3 calls (iii) atleast 5 calls
10. The distribution of the number of road accidents per day in a city is poisson with mean 4. Find the number of days out of 100 days when there will be (i) no accident (ii) atleast 2 accidents and (iii) at most 3 accidents.
11. Assuming that a fatal accident in a factory during the year is 1/1200, calculate the probability that in a factory employing 300 workers there will be atleast two fatal accidents in a year. (given $e^{-0.25} = 0.7788$)
12. The average number of customers, who appear in a counter of a certain bank per minute is two. Find the probability that during a given minute (i) No customer appears (ii) three or more customers appear .

Continuous distribtuion

The binomial and Poisson distributions discussed in the previous chapters are the most useful theoretical distributions for discrete variables. In order to have mathematical distributions suitable for dealing with quantities whose magnitudes vary continuously like weight, heights of individual, a continuous distribution is needed. Normal distribution is one of the most widely used continuous distribution.

Normal distribution is the most important and powerful of all the distribution in statistics. It was first introduced by De Moivre in 1733 in the development of probability. Laplace (1749-1827) and Gauss (1827-1855) were also associated with the development of Normal distribution.

7.1.3 Normal Ditribution

Definition 7.3

A random variable X is said to follow a normal distribution with parameters mean μ and variance σ^2, if its probability density function is given by

$$f(x:\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$$

$$\left.\begin{array}{r} -\infty < x < \infty, \\ -\infty < \alpha < \infty, \\ \sigma > 0 \end{array}\right\}$$

Normal distribution is diagrammatically represented as follows :

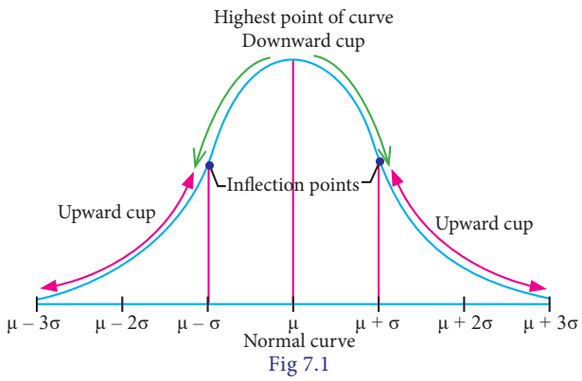


Fig 7.1

Normal distribution is a limiting case of binomial distribution under the following conditions:

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